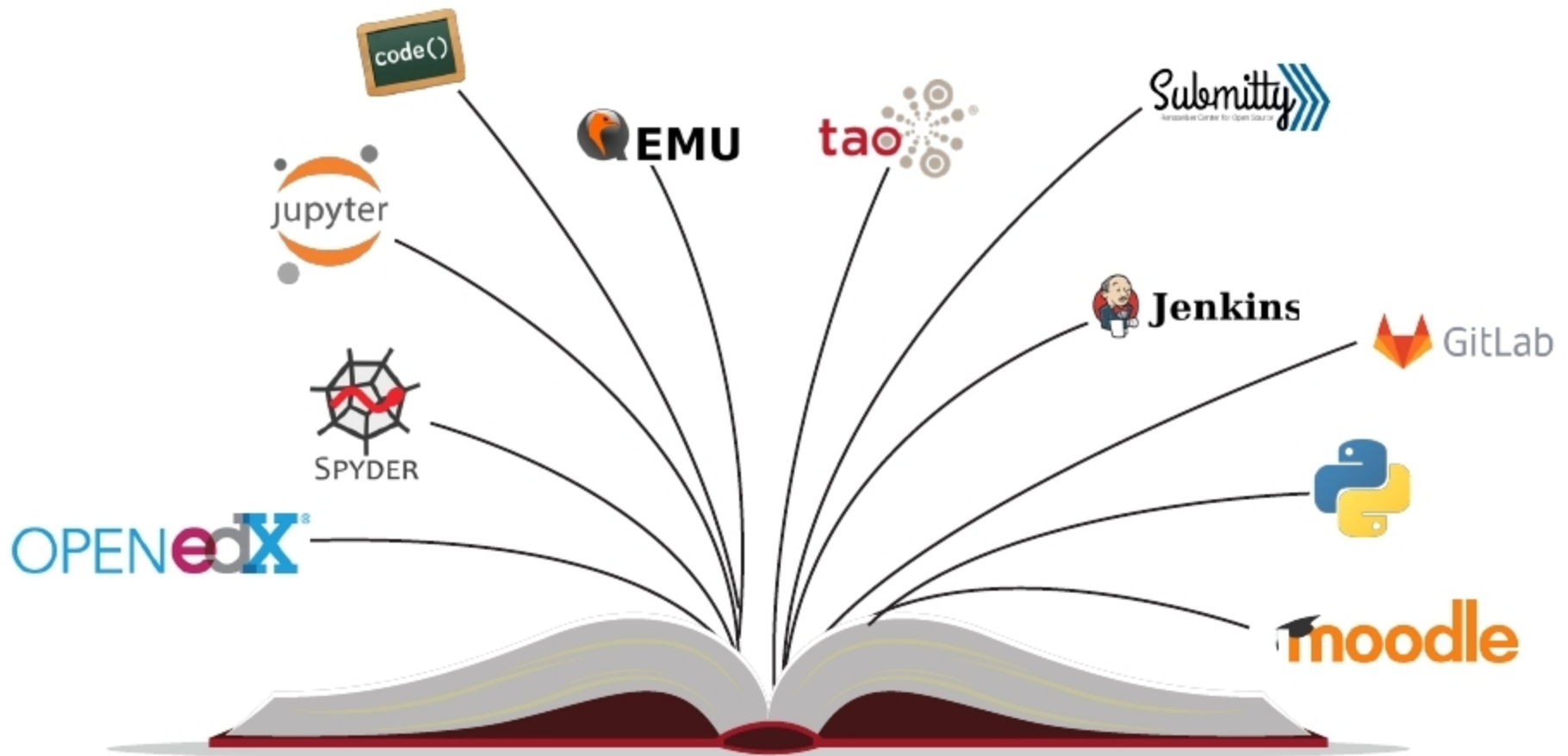


The Role of Open Source Software in Engineering Education



Open source software can play a vital role in education, especially when it comes to the engineering domain. As engineering involves complex tools and processes to deliver solutions, OSS ensures lower costs and more robust code built on open standards that can be continually peer-reviewed. This helps in effective teaching and the management of academics.

The most important open source software for an institution that is educating engineers are:

- Learning management systems (LMS) and tools
- Code collaboration tools
- Submission tracking and assessment frameworks
- Circuit/MCU simulation tools
- Interactive notebooks and IDEs (integrated development environments)

Though we are talking about engineering education here, many of the open source tools and frameworks

discussed in this article are equally applicable to other academic and corporate segments.

Learning management systems (LMS) and tools

Open edX

This is a digital MOOC (massive open online course) framework created by MIT and Harvard, with all the necessary functionalities to deliver and manage content. It has had at least one major release per year ever since its launch in 2013, and has been used by more than 35+ million people.

The LMS is based on the Django application framework with Python for the back-end, while Mako, JavaScript and CSS are primarily used for the front-end. MongoDB and MySQL have been used for the database. Multiple interfaces are enabled for external application through LTI (learning tool interoperability) and the edX Xblocks API. The course creation is done in OLXML and is interoperable with easy export and import. The data analytics engine gathers the events and makes insights available with the help of Hadoop and the REST API. Multiple educational institutions and corporates